



Inverse Data Envelopment Analysis-Based Decision Support for Road Safety-Oriented Urban and Regional Transport Planning and Infrastructure Management

Miroslav Đerić^{1*}, Marko Subotić², Rasa Ušpalytė-Vitkūnienė³, Boris Gitolendia⁴

¹ Ministry of Communications and Transport of Bosnia and Herzegovina, 71000 Sarajevo, Bosnia and Herzegovina

² Faculty of Transport and Traffic Engineering, University of East Sarajevo, 74000 Dobo, Bosnia and Herzegovina

³ Road Department, Vilnius Gediminas Technical University, LT-10224 Vilnius, Lithuania

⁴ Faculty of Transport Systems and Mechanical Engineering, Georgian Technical University, 0160 Tbilisi, Georgia

* Correspondence: Miroslav Đerić (miroslav.djeric@mkt.gov.ba)

Received: 07-02-2026

Revised: 08-14-2026

Accepted: 08-18-2026

Citation: M. Đerić, M. Subotić, R. Ušpalytė-Vitkūnienė, and B. Gitolendia, “Inverse data envelopment analysis-based decision support for road safety-oriented urban and regional transport planning and infrastructure management,” *J. Urban Dev. Manag.*, vol. 5, no. 3, pp. 225–239, 2026. <https://doi.org/10.56578/judm050304>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Urban and regional transport authorities need reliable methods to identify road sections where traffic exposure and crash outcomes indicate comparatively unfavorable safety performance, particularly when planning and infrastructure-management resources are limited. This study investigates how inverse data envelopment analysis (DEA) can support the comparative assessment and prioritization of two-lane road sections for road safety-oriented transport planning and infrastructure management. Twelve main road sections within the jurisdiction of the East Sarajevo Police Administration were analyzed using data from 2019–2023. Annual average daily traffic (AADT) was used to represent traffic exposure, while the average annual numbers of fatal, serious-injury, and slight-injury crashes were used as adverse safety outcomes. A Charnes–Cooper–Rhodes (CCR) model-based inverse DEA framework was applied to determine the relative position of each road section against an empirically defined efficiency frontier. A constrained radial scenario was then used to estimate reference values for the three crash-severity indicators while keeping AADT unchanged. The results showed that two road sections formed the relative efficiency frontier, while substantial differences were observed among the remaining sections. The two least favorable sections deviated from the reference level by more than 91%, whereas another section was only 2.40% below the frontier. These findings indicate that the proposed framework can distinguish between road sections requiring different levels of planning attention rather than relying on a simple efficient–inefficient classification. The approach provides a transparent decision-support tool for urban and regional transport planning, helping authorities screen road sections, direct detailed road-safety assessments, and support the sequencing of maintenance, reconstruction, and infrastructure-improvement activities.

Keywords: Road safety; Inverse data envelopment analysis; Urban and regional transport planning; Infrastructure management; Road-section prioritization; Two-lane roads

1 Introduction

Road transport is a fundamental component of urban and regional transport systems, connecting settlements, economic activities, and different parts of a territory. Two-lane roads are particularly important in this context because they account for a substantial share of road networks and often provide essential connections between urban areas, smaller settlements, and rural communities. At the same time, these roads generally carry two-way traffic without physical separation between opposing flows, offer limited overtaking opportunities, and are strongly affected by road geometry and local conditions. Their safety performance is therefore an important concern not only for road-safety management but also for urban and regional transport planning and infrastructure management. Road sections may differ considerably in traffic volume, crash frequency, and crash severity, making comparisons based on a single indicator potentially misleading. Planning and infrastructure-management decisions consequently require an analytical approach that considers traffic exposure and different crash outcomes together and identifies sections with comparatively unfavorable safety performance.

Data envelopment analysis (DEA) provides a suitable basis for such comparisons because it allows homogeneous decision-making units (DMUs) to be evaluated simultaneously against multiple indicators without requiring a predefined functional relationship among them. Banker et al. [1] developed the Banker–Charnes–Cooper (BCC) extension of DEA, while classical DEA models are generally used to establish the relative position of an observed unit against an efficiency frontier constructed from the available data. In a road-safety context, this type of benchmarking can distinguish sections with relatively favorable combinations of traffic exposure and safety outcomes from those requiring closer examination. The present study focuses on the inverse DEA approach, which goes beyond identifying the current relative position of a unit by examining the indicator values associated with a selected efficiency level. This additional step is relevant to planning because it provides information not only on relative performance but also on the magnitude of the deviation from a defined reference level.

From an urban and regional transport-planning perspective, identifying a road section with comparatively poor safety performance is only the first stage of the decision process. Authorities responsible for road networks also need to understand how far individual sections lie from an appropriate reference level so that detailed safety assessments, maintenance, reconstruction, and possible infrastructure improvements can be considered in a more systematic sequence. Wei et al. [2] formalised the concept of inverse DEA by examining the input or output values required after changes in the model while maintaining a given level of relative efficiency. Later studies extended the approach to cases in which a target efficiency level is specified. Inverse DEA can therefore link the current relative position of a unit to reference values that indicate the scale of change associated with reaching a selected level. This capability is particularly relevant to road infrastructure management, where decision makers need to know not only which sections perform less favorably, but also whether the relative deviations among those sections are small or substantial.

Despite this potential, a practical gap remains between road-safety benchmarking and the information required for infrastructure planning and management. A simple classification of road sections as relatively efficient or inefficient does not reveal whether sections below the frontier have similar or markedly different distances from the reference level. This distinction matters when road authorities must decide where more detailed professional analysis should be directed and how limited analytical and financial resources can be allocated before specific interventions are selected. The research problem addressed in this paper is therefore to use an inverse DEA model to compare the relative road-safety performance of road sections, identify sections with less favorable results, and quantify their deviation from a defined reference level. The model is also used to estimate reference changes in the selected safety indicators for a specified level of relative efficiency. These reference values are intended as decision-support information rather than forecasts of future crash numbers or direct prescriptions for engineering interventions.

Accordingly, the main aim of this study is to develop a methodological framework for assessing the relative road-safety performance of two-lane road sections within the jurisdiction of the East Sarajevo Police Administration and to examine how the resulting information can support urban and regional transport planning and road infrastructure management. The framework identifies reference sections, distinguishes sections below the relative efficiency frontier, and estimates the scale of change in the selected indicators associated with reaching the reference level. Its contribution lies in connecting road-safety benchmarking with a reference-based assessment of the magnitude of relative deviation, thereby providing more differentiated information than a simple frontier classification. The results can support the systematic screening and prioritization of road sections for further safety analysis and provide an additional analytical basis for planning the sequence of maintenance, reconstruction, and infrastructure-improvement activities. The framework is intended to complement, rather than replace, detailed engineering assessment when specific road-safety interventions are considered.

2 Literature Review

DEA is a non-parametric mathematical programming method used to assess the relative efficiency of comparable DMUs with multiple inputs and outputs. The Charnes–Cooper–Rhodes (CCR) model was introduced by Charnes et al. [3], while Banker et al. [1] later developed the BCC model, which separates technical efficiency from scale efficiency. DEA has since been applied to a range of transport-planning and infrastructure-management problems in which performance cannot be represented adequately by a single indicator. Odeck [4], for example, used DEA and Malmquist indices to analyse vehicle inspection services and later applied the method to road-safety benchmarking [5]. Ozbek et al. [6] examined data and model-specification issues in road-maintenance efficiency studies, while Fancello et al. [7] used DEA to assess the performance of urban road systems. Together, these applications show the usefulness of DEA for comparing transport and infrastructure units whose performance depends on several interacting dimensions.

In road-safety research, DEA is particularly useful because traffic exposure, safety outcomes, and other risk-related indicators can be considered within a common analytical framework. Hermans et al. [8] used DEA to benchmark road safety across European countries and identify reference units, while Shen et al. [9] linked DEA assessment with risk evaluation and target setting. Van Espen [10] combined DEA with crash-prediction modelling in a national road-safety assessment, and Nikolaou and Dimitriou [11] used DEA and cross-efficiency analysis to compare road-safety policies across European countries over several years. More recently, Shen et al. [12] proposed a benchmarking framework

that emphasises reference units, structured indicators, and the use of results in road-safety management. These studies demonstrate that DEA can provide a comparative basis for safety assessment, although the interpretation and practical use of the resulting efficiency measures depend on the spatial and decision-making context in which the analysis is undertaken.

DEA has also been applied at regional and local levels, which is particularly relevant to the use of road-safety indicators in urban and regional transport planning. Alper et al. [13] assessed the efficiency of local municipalities in providing road safety by combining resource, safety, and outcome indicators. Antić et al. [14] compared regional road-safety performance and showed that DEA can distinguish relatively better-performing territorial units. Egilmez and McAvoy [15] used a DEA-Malmquist approach to compare U.S. states in terms of fatality reduction, while Bastos et al. [16] developed a composite DEA indicator for Brazilian states. Xu et al. [17] applied a two-stage DEA model to provincial road-traffic-safety performance in China. Collectively, this body of research indicates that DEA-based benchmarking can operate across different spatial scales and can provide information relevant to authorities responsible for territorial transport systems and road-safety management.

Other studies have moved from territorial comparisons to the assessment of individual road sections and the prioritization of subsequent interventions. Mitrović Simić et al. [18] combined DEA with multi-criteria decision-making methods to evaluate road-section safety using geometric road parameters, while Sadeghi and Mohammadzadeh Moghaddam [19] used DEA to support the prioritisation of road-safety projects under uncertainty. For two-lane roads, traffic exposure and geometric characteristics are especially important. Cafiso et al. [20] showed the value of combining exposure, geometry, design consistency, and contextual variables in crash models for two-lane rural roads. Montella [21] also showed that hotspot-identification results can depend on the selected method and criteria, which is relevant when DEA results are interpreted. These findings underline the need to treat road-section benchmarking as part of a broader planning and engineering process rather than as a stand-alone basis for selecting interventions.

Recent studies often combine DEA with other analytical methods because road safety is a multidimensional problem. Damjanović et al. [22] analysed the influence of the number of vehicles through a multiphase model, while Željko et al. [23] proposed another multiphase approach for road-safety evaluation. These studies indicate that DEA results are best interpreted together with other indicators and additional validation procedures. In this sense, DEA is primarily a screening and benchmarking tool: it can identify units that require closer attention, but it cannot replace an engineering analysis of crash causes. This distinction is particularly important for planning and infrastructure management because a relative performance measure can indicate where further analysis may be warranted without determining which physical, operational, or management intervention should ultimately be implemented.

Inverse DEA extends the classical DEA approach by examining what changes in the indicators correspond to a given efficiency level. Wei et al. [2] developed one of the basic inverse DEA models for estimating required input and output values. Gattoufi et al. [24] later demonstrated further applications in scenario analysis, while Emrouznejad et al. [25] reviewed the main developments and variants of the method. This logic is particularly relevant to road-section analysis because the current relative position of a section can be supplemented by an estimate of how much selected indicators would need to change to reach a reference level. In a planning context, this makes it possible to distinguish between road sections that are only marginally below the reference frontier and those that show much larger relative deviations.

The literature therefore provides a well-established basis for DEA-based road-safety benchmarking, including applications across countries, regions, municipalities, and individual road sections. A less developed aspect is the use of inverse DEA to translate the relative position of individual road sections into explicit reference values that can support the initial stages of transport planning and infrastructure-management prioritization. Addressing this gap does not imply that model-derived reference values can predict future crash reductions or prescribe particular infrastructure measures. Rather, their value lies in adding a measure of relative distance to conventional benchmarking results. This study adopts that perspective by combining the assessment of current road-section performance with a constrained inverse DEA scenario in which traffic exposure remains unchanged while adverse road-safety outcomes are projected toward a common reference level. The resulting information provides a basis for identifying sections that warrant more detailed investigation and for supporting subsequent planning decisions, while detailed engineering analysis remains necessary before any intervention is selected.

3 Research Methodology

3.1 Research Concept

This study develops an inverse DEA-based framework for assessing the relative road-safety performance of two-lane main road sections within the jurisdiction of the East Sarajevo Police Administration. The analysis is designed to support the screening and prioritization of road sections for further safety assessment within urban and regional transport planning and road infrastructure management. Because the analysed sections differ in both traffic exposure and crash outcomes, comparisons based on a single indicator may provide an incomplete representation of their relative safety performance. The proposed framework therefore considers traffic intensity and multiple

crash-severity indicators simultaneously and evaluates each road section relative to the performance observed across the analysed network.

The methodology consists of three connected stages. The first stage involves database development, including the processing of road-crash data, estimation of missing annual average daily traffic (AADT) values, and preparation of the final indicator matrix. In the second stage, a CCR model with a reversed indicator configuration is applied to assess the relative position of the 12 road sections and identify those forming the empirical reference frontier. The third stage applies a constrained radial inverse DEA scenario to determine reference values for the safety indicators and quantify the relative deviation of the remaining sections from the selected reference level. The resulting framework therefore provides two complementary pieces of information: the current relative position of each road section and the magnitude of its deviation from the reference level. Together, these outputs provide a systematic basis for identifying sections that warrant more detailed road-safety analysis and for informing subsequent transport-planning and infrastructure-management decisions. The three-stage research workflow is presented in Figure 1.

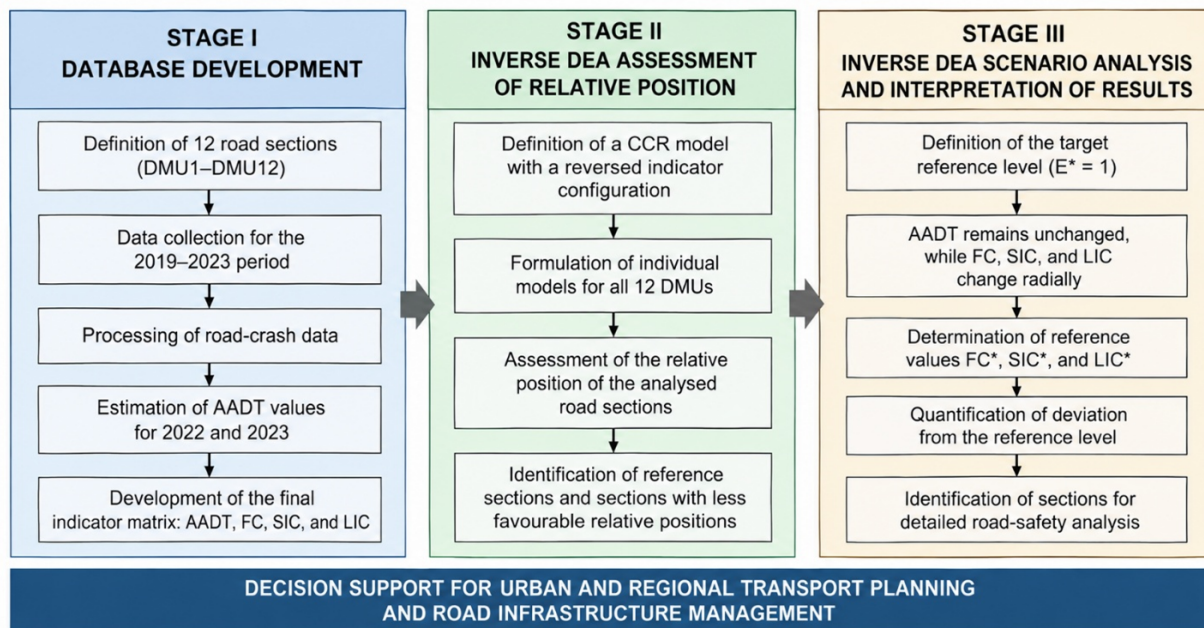


Figure 1. Three-stage research workflow based on the inverse DEA framework

Note: DEA = data envelopment analysis; CCR = Charnes–Cooper–Rhodes model; DMUs = decision-making units; AADT = annual average daily traffic; FC = fatal crashes; SIC = serious-injury crashes; LIC = slight-injury crashes; E^* = target reference efficiency level; FC^* , SIC^* , and LIC^* = reference values of fatal, serious-injury, and slight-injury crashes obtained from the inverse DEA scenario analysis.

The study covers 12 main road sections, each treated as a homogeneous DMU within the analytical framework. This common unit of analysis allows the sections to be compared within the same model using a consistent set of indicators. The database contains road-section length, AADT, and crash data for the 2019–2023 period.

3.2 Data Collection and Processing

The database was developed from road-section characteristics, traffic-volume records, and official road-crash data. Crash data were obtained from the Ministry of the Interior of Republika Srpska, Police Directorate, for the area under the jurisdiction of the East Sarajevo Police Administration and were provided on 1 August 2025. The dataset covers the 2019–2023 period and records crashes according to three severity categories: fatal, serious-injury, and slight-injury crashes. The crash records were organised by road section and converted into average annual values to ensure a consistent basis for comparison in the inverse DEA analysis. Road-section length and AADT data were also collected for each section. These data were subsequently used to construct the indicator set described in Section 3.4.

3.3 Estimation of Annual Average Daily Traffic Values

Available AADT records were used to represent traffic exposure. Complete AADT observations for 2022 and 2023 were not available for all road sections; therefore, the missing values were estimated separately for each section using linear regression. The year 2020 was excluded from the estimation because traffic volumes during the COVID-19 pandemic differed considerably from the patterns observed before and after that year and could therefore distort the underlying traffic trend. The regression models were estimated using the available observations for 2017, 2018, 2019, and 2021.

A separate linear regression equation and coefficient of determination (R^2) were obtained for each road section. The fitted regression lines were then extrapolated to estimate the missing AADT values for 2022 and 2023. These estimated values were combined with the available observations for the remaining years to calculate the average AADT for each road section over the 2019–2023 analysis period. The estimated values were used only to complete the traffic-exposure series required for the comparative analysis and were not treated as forecasts of future traffic conditions.

3.4 Definition of Model Indicators

Each of the 12 main road sections was treated as a separate DMU. The indicator structure was designed to compare traffic exposure with recorded crash outcomes across the analysed sections. AADT was used to represent traffic volume and exposure, while the average annual numbers of fatal crashes (FC), serious-injury crashes (SIC), and slight-injury crashes (LIC) represented adverse road-safety outcomes.

Road-section length was considered during the preliminary analysis but was not included as an indicator in the final DEA specification. It was retained as a descriptive characteristic of the analysed sections because the model focuses specifically on the relationship between traffic exposure and recorded crash outcomes. The final analytical structure therefore contains four indicators: AADT, FC, SIC, and LIC.

For the formal DEA classification presented in Table 1, AADT is defined as the input because it represents the level of traffic exposure associated with each road section, whereas FC, SIC, and LIC are formally classified as outputs representing observed safety outcomes. These outputs are undesirable rather than beneficial: lower crash values indicate a more favorable safety condition for a given level of traffic exposure. This distinction is important for interpreting the model. As explained in Section 3.5, the indicators are subsequently arranged in a reversed configuration in the CCR formulation so that a more favorable relationship between traffic exposure and adverse crash outcomes corresponds to a higher relative-performance value.

Table 1. Indicators and their role in the model

Symbol	Input/Output	Indicator	Role in the Model
AADT	Input	Annual average daily traffic (AADT) (vehicles/day)	Traffic volume/exposure
FC	Output	Average annual number of fatal crashes (FC)	Adverse road-safety outcome
SIC	Output	Average annual number of serious-injury crashes (SIC)	Adverse road-safety outcome
LIC	Output	Average annual number of slight-injury crashes (LIC)	Adverse road-safety outcome

Table 2 reports the descriptive characteristics and values of the selected indicators for the 12 road sections included in the analysis. Road-section length is reported for contextual comparison but is not included in the final DEA model.

Table 2. Indicator values for the analysed road sections

DMUs	Road Section	Length (km)	AADT	FC	SIC	LIC
DMU1	Kula–Krupac	5.165	7,644	0.40	4.40	1.00
DMU2	Podromanija–Sumbulovac	21.559	3,911	0.20	7.20	2.20
DMU3	Sumbulovac–Ljubogošta	5.030	5,549	0.40	3.80	1.80
DMU4	Ljubogošta–Pale 1	4.460	5,277	0.20	3.00	1.60
DMU5	Pale 1–Podgrab	15.660	1,928	0.80	2.80	1.20
DMU6	Podgrab–RS/FBiH Border (Prača)	6.155	1,794	0.00	0.40	0.20
DMU7	Krupac–RS/FBiH Border (Bogatići)	11.265	3,306	0.00	0.60	0.40
DMU8	Trnovo–Dobro Polje	12.021	1,751	0.00	0.80	0.20
DMU9	Sokolac–Podromanija	3.598	7,282	0.40	2.00	1.20
DMU10	Vlasenica–Han Pijesak 1	20.317	2,214	0.00	2.40	0.80
DMU11	Han Pijesak 2–Sokolac	27.483	1,932	0.20	5.20	2.40
DMU12	Podromanija–Rogatica	28.598	2,730	0.40	10.40	3.40

Note: DMUs = decision-making units; AADT = annual average daily traffic; FC = fatal crashes; SIC = serious-injury crashes; LIC = slight-injury crashes; RS = Republic of Srpska; FBiH = Federation of Bosnia and Herzegovina.

Figure 2 compares the length of each road section with the total number of crashes recorded during the study period. Podromanija–Rogatica (DMU12) is the longest section, at 28.598 km, whereas Sokolac–Podromanija (DMU9)

is the shortest, at 3.598 km. The figure provides descriptive context for the variation in section characteristics but does not imply a direct relationship between section length and crash occurrence.

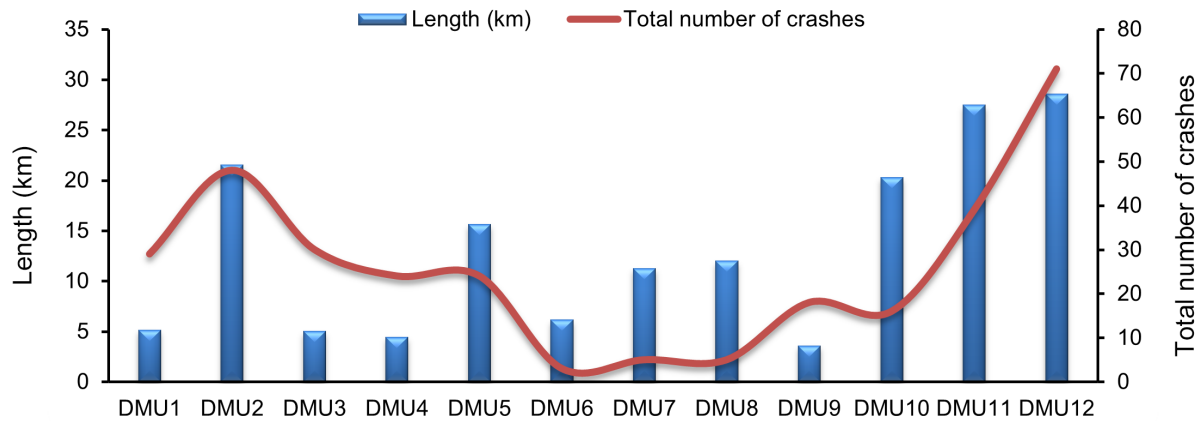


Figure 2. Road-section lengths and total number of road crashes
Note: DMU = decision-making unit.

Figure 3 presents the AADT values used to represent traffic exposure in the model. Kula–Krupac (DMU1) recorded the highest AADT, at 7,644 vehicles/day, followed by Sokolac–Podromanija (DMU9), at 7,282 vehicles/day. The lowest AADT was recorded on Trnovo–Dobro Polje (DMU8), at 1,751 vehicles/day.

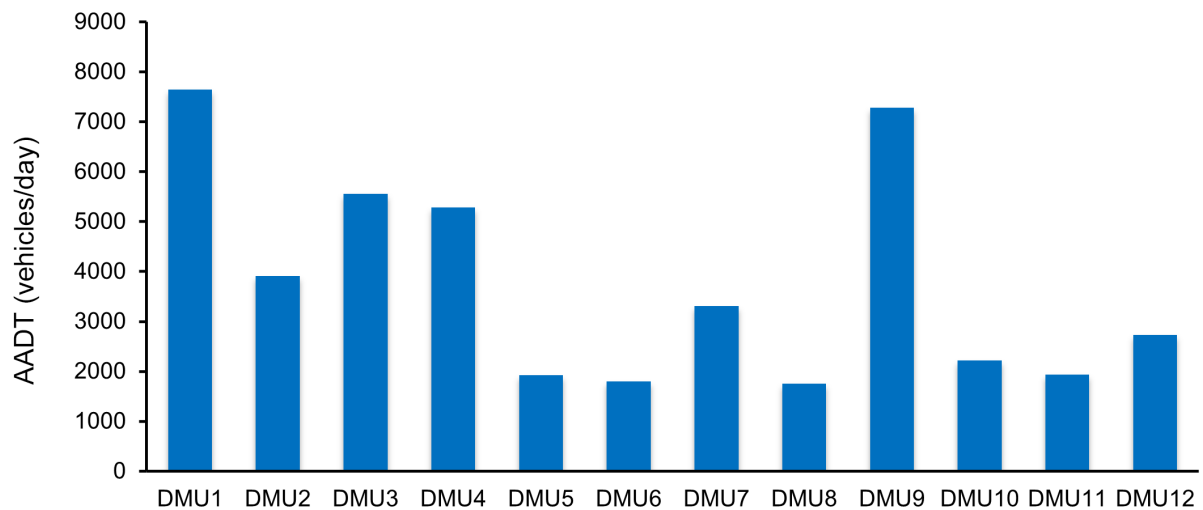


Figure 3. Average AADT on the analysed road sections over the 2019–2023 period
Note: AADT = annual average daily traffic; DMU = decision-making unit.

Figure 4 presents the distribution of crash outcomes by severity over the five-year study period. Podromanija–Rogatica (DMU12) recorded the highest number of serious-injury crashes (52 cases), while no fatal crashes were recorded on Podgrab–RS/FBiH Border (Prača) (DMU6), Krupac–RS/FBiH Border (Bogatići) (DMU7), Trnovo–Dobro Polje (DMU8), or Vlasenica–Han Pijesak 1 (DMU10) during 2019–2023.

Figure 5 presents the average annual numbers of FC, SIC, and LIC used in the inverse DEA analysis. Podromanija–Rogatica (DMU12) had the highest average number of serious-injury crashes, with 10.40 cases per year. No fatal crashes were recorded on Podgrab–RS/FBiH Border (Prača) (DMU6), Krupac–RS/FBiH Border (Bogatići) (DMU7), Trnovo–Dobro Polje (DMU8), or Vlasenica–Han Pijesak 1 (DMU10) during the study period.

3.5 Formulation of the Inverse Data Envelopment Analysis Model

The model uses a CCR specification with constant returns to scale and is written in multiplier form. The indicators are arranged in the reverse manner of their formal input-output classification presented in Table 1. AADT, formally defined as the input and used here as a measure of traffic exposure, is included in the numerator. FC, SIC, and LIC, formally defined as outputs and representing adverse road-safety outcomes, are included in the denominator. This reversed arrangement is used to assess the relative road-safety position of the analysed sections. The weighting

coefficients w_1 , w_2 , w_3 , and w_4 are determined separately for each DMU.

$$E_o = \frac{AADT_o \cdot w_1}{FC_o \cdot w_2 + SIC_o \cdot w_3 + LIC_o \cdot w_4} \quad (1)$$

The fractional expression is converted into a linear programming model by normalising the denominator of the observed DMU to 1:

$$\begin{aligned} \max \quad & E_o = AADT_o \cdot w_1 \\ \text{s.t.} \quad & AADT_j \cdot w_1 - (FC_j \cdot w_2 + SIC_j \cdot w_3 + LIC_j \cdot w_4) \leq 0, \quad j = 1, \dots, 12 \\ & FC_o \cdot w_2 + SIC_o \cdot w_3 + LIC_o \cdot w_4 = 1 \\ & w_1, w_2, w_3, w_4 > 0 \end{aligned} \quad (2)$$

The constraints ensure that, for a given set of weights, no analysed road section obtains a weighted efficiency ratio greater than one. The normalisation condition fixes the denominator of the observed DMU to 1 and converts the fractional CCR expression into a linear programme. A separate optimisation is performed for each DMU, with positive weights as specified in the LINGO formulations. A value of $E_o = 1$ places a section on the relative efficiency frontier, while $E_o < 1$ indicates a less favourable relative relationship between traffic exposure and recorded crash outcomes. All models were solved separately in LINGO.

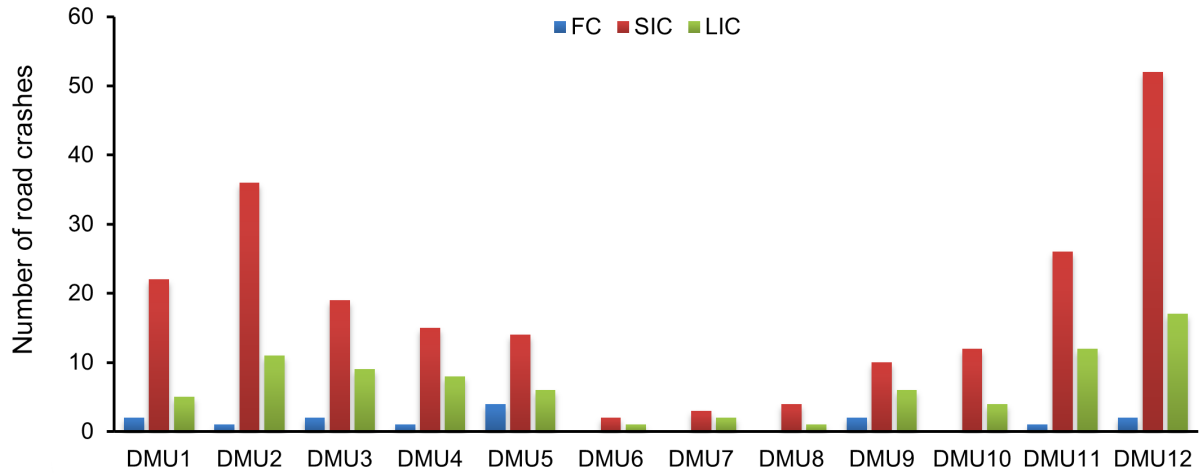


Figure 4. Total number of road crashes by severity on the analysed road sections, 2019–2023

Note: DMU = decision-making unit; FC = fatal crashes; SIC = serious-injury crashes; LIC = slight-injury crashes.

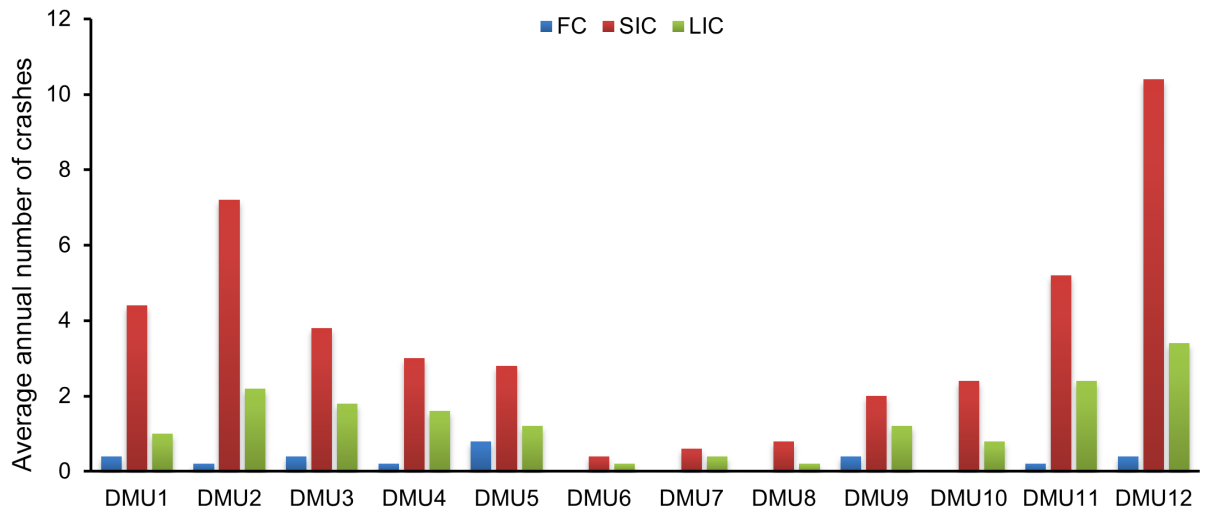


Figure 5. Average annual number of road crashes by severity on the analysed road sections, 2019–2023

Note: DMU = decision-making unit; FC = fatal crashes; SIC = serious-injury crashes; LIC = slight-injury crashes.

3.5.1 Example of the inverse data envelopment analysis model for decision-making unit 1–Kula–Krupac

For Kula–Krupac (DMU1), the average AADT is 7,644 vehicles/day. The corresponding average annual crash values are FC = 0.40, SIC = 4.40, and LIC = 1.00. The objective function and model constraints are given below:

$$\begin{aligned}
 \max \quad & E_1 = 7644w_1 \\
 & 7644w_1 - (0.4w_2 + 4.4w_3 + 1.0w_4) \leq 0 \\
 & 3911w_1 - (0.2w_2 + 7.2w_3 + 2.2w_4) \leq 0 \\
 & 5549w_1 - (0.4w_2 + 3.8w_3 + 1.8w_4) \leq 0 \\
 & 5277w_1 - (0.2w_2 + 3.0w_3 + 1.6w_4) \leq 0 \\
 & 1928w_1 - (0.8w_2 + 2.8w_3 + 1.2w_4) \leq 0 \\
 & 1794w_1 - (0.0w_2 + 0.4w_3 + 0.2w_4) \leq 0 \\
 & 3306w_1 - (0.0w_2 + 0.6w_3 + 0.4w_4) \leq 0 \\
 & 1751w_1 - (0.0w_2 + 0.8w_3 + 0.2w_4) \leq 0 \\
 & 7282w_1 - (0.4w_2 + 2.0w_3 + 1.2w_4) \leq 0 \\
 & 2214w_1 - (0.0w_2 + 2.4w_3 + 0.8w_4) \leq 0 \\
 & 1932w_1 - (0.2w_2 + 5.2w_3 + 2.4w_4) \leq 0 \\
 & 2730w_1 - (0.4w_2 + 10.4w_3 + 3.4w_4) \leq 0 \\
 & 0.4w_2 + 4.4w_3 + 1.0w_4 = 1 \\
 & w_1, w_2, w_3, w_4 > 0
 \end{aligned} \tag{3}$$

The full formulation is shown for Kula–Krupac (DMU1) to illustrate the model structure. The remaining 11 models are constructed in the same way. The 12 inequality constraints are common to all models, while the objective function and the normalisation constraint change according to the DMU being evaluated.

3.5.2 Individualisation of the inverse data envelopment analysis model for all 12 decision-making units

Table 3 summarises the objective function and normalisation constraint for each of the 12 DMUs. The common inequality constraints are not repeated.

Table 3. Objective function and normalisation constraint of the individual inverse DEA models

DMUs	Road Section	Objective Function	Normalisation Constraint
DMU1	Kula–Krupac	$\max = 7644w_1$	$0.4w_2 + 4.4w_3 + 1.0w_4 = 1$
DMU2	Podromanija–Sumbulovac	$\max = 3911w_1$	$0.2w_2 + 7.2w_3 + 2.2w_4 = 1$
DMU3	Sumbulovac–Ljubogošta	$\max = 5549w_1$	$0.4w_2 + 3.8w_3 + 1.8w_4 = 1$
DMU4	Ljubogošta–Pale 1	$\max = 5277w_1$	$0.2w_2 + 3.0w_3 + 1.6w_4 = 1$
DMU5	Pale 1–Podgrab	$\max = 1928w_1$	$0.8w_2 + 2.8w_3 + 1.2w_4 = 1$
DMU6	Podgrab–RS/FBiH Border (Prača)	$\max = 1794w_1$	$0.0w_2 + 0.4w_3 + 0.2w_4 = 1$
DMU7	Krupac–RS/FBiH Border (Bogatići)	$\max = 3306w_1$	$0.0w_2 + 0.6w_3 + 0.4w_4 = 1$
DMU8	Trnovo–Dobro Polje	$\max = 1751w_1$	$0.0w_2 + 0.8w_3 + 0.2w_4 = 1$
DMU9	Sokolac–Podromanija	$\max = 7282w_1$	$0.4w_2 + 2.0w_3 + 1.2w_4 = 1$
DMU10	Vlasenica–Han Pijesak 1	$\max = 2214w_1$	$0.0w_2 + 2.4w_3 + 0.8w_4 = 1$
DMU11	Han Pijesak 2–Sokolac	$\max = 1932w_1$	$0.2w_2 + 5.2w_3 + 2.4w_4 = 1$
DMU12	Podromanija–Rogatica	$\max = 2730w_1$	$0.4w_2 + 10.4w_3 + 3.4w_4 = 1$

Note: DEA = data envelopment analysis; DMUs = decision-making units; RS = Republic of Srpska; FBiH = Federation of Bosnia and Herzegovina. The 12 inequalities that limit the weighted ratios are identical across all models and are listed in the DMU1 example. Therefore, this table reports only the objective function and normalisation constraint that vary among DMUs.

3.6 Inverse Data Envelopment Analysis Scenario Analysis

After the relative positions were obtained, a scenario analysis was carried out for the sections below the relative efficiency frontier. A target efficiency level E_o^* was specified, while AADT was kept at its observed value. FC, SIC, and LIC were then changed proportionally. This produced a single proportional-change factor for the three adverse road-safety outcomes and the corresponding reference values for the selected target level. For a DMU_o below the relative efficiency frontier, with an initial efficiency value E_o , the target level E_o^* is defined as:

$$E_o < E_o^* \leq 1 \tag{4}$$

$$\alpha_o = \frac{E_o}{E_o^*} \quad (5)$$

$$FC_o^* = \alpha_o \cdot FC_o, \quad SIC_o^* = \alpha_o \cdot SIC_o, \quad LIC_o^* = \alpha_o \cdot LIC_o, \quad AADT_o^* = AADT_o \quad (6)$$

$$\Delta FC_o = FC_o - FC_o^*, \quad \Delta SIC_o = SIC_o - SIC_o^*, \quad \Delta LIC_o = LIC_o - LIC_o^* \quad (7)$$

When the target is set at $E_o^* = 1$, the radial factor becomes $\alpha_o = E_o$. The three adverse road-safety outcomes are therefore adjusted proportionally by the same factor, while AADT remains unchanged. This represents a constrained radial projection towards the existing CCR-based relative efficiency frontier, rather than an unconstrained Inverse DEA formulation where each indicator is allowed to vary independently. The restriction was introduced to make the comparison transparent and to express the required change as a single percentage value. The resulting values are model-based reference values and should not be interpreted as forecasts, mandatory targets, or directly prescribed infrastructure measures.

3.6.1 Example of determining reference values for decision-making unit 1–Kula–Krupac

The calculation for Kula–Krupac (DMU1) is used to demonstrate how the reference values are obtained. The initial efficiency value for this section is $E_1 = 0.8522$. With the reference level set at $E_1^* = 1$, Eq. (5) gives $\alpha_1 = 0.8522$. Substituting this value into Eq. (6) results in the following reference values:

$$FC_1^* = 0.40 \cdot 0.8522 = 0.341$$

$$SIC_1^* = 4.40 \cdot 0.8522 = 3.750$$

$$LIC_1^* = 1.00 \cdot 0.8522 = 0.852$$

With AADT kept at 7,644 vehicles/day, the scenario corresponds to a proportional adjustment of about 14.78% in FC, SIC, and LIC. These values do not represent forecasts of future crash numbers; instead, they are reference values generated by the selected inverse DEA scenario.

4 Results and Discussion

The analysis first determined the relative position of each road section against an empirical frontier defined by the most favorable observed relationships between AADT and crash outcomes. The subsequent scenario analysis quantified the relative distance between sections below the frontier and the selected reference level. The E value and the radial-reduction percentage therefore describe two related aspects of the same relative position: the former indicates where a section stands within the analysed network, while the latter expresses the magnitude of its deviation from the reference frontier. Under the adopted scenario, AADT was held constant and FC, SIC, and LIC were changed proportionally. The resulting values should therefore be interpreted as model-based reference projections rather than forecasts of future crash numbers or formal safety-reduction targets.

4.1 Results of the Relative Road-Safety Performance Assessment

Table 4 presents the relative road-safety performance of the 12 analysed sections. For the selected indicators and the observed set of DMUs, $E = 1$ indicates that a section lies on the CCR efficiency frontier. Values below one represent a less favorable relative relationship between traffic exposure, measured by AADT, and the recorded crash outcomes.

Table 4. Results of the relative road-safety performance assessment of the road sections

DMUs	Road Section	E	Status
DMU1	Kula–Krupac	0.8522	Below the relative efficiency frontier
DMU2	Podromanija–Sumbulovac	0.1982	Below the relative efficiency frontier
DMU3	Sumbulovac–Ljubogošta	0.3437	Below the relative efficiency frontier
DMU4	Ljubogošta–Pale 1	0.3751	Below the relative efficiency frontier
DMU5	Pale 1–Podgrab	0.1791	Below the relative efficiency frontier
DMU6	Podgrab–RS/FBiH Border (Prača)	1.0000	On the relative efficiency frontier
DMU7	Krupac–RS/FBiH Border (Bogatići)	1.0000	On the relative efficiency frontier
DMU8	Trnovo–Dobro Polje	0.9760	Below the relative efficiency frontier
DMU9	Sokolac–Podromanija	0.7139	Below the relative efficiency frontier
DMU10	Vlasenica–Han Pijesak 1	0.3085	Below the relative efficiency frontier
DMU11	Han Pijesak 2–Sokolac	0.0897	Below the relative efficiency frontier
DMU12	Podromanija–Rogatica	0.0895	Below the relative efficiency frontier

Note: DMUs = decision-making units; E = relative efficiency score; RS = Republic of Srpska; FBiH = Federation of Bosnia and Herzegovina.

Two of the 12 road sections, Podgrab–RS/FBiH Border (Prača) (DMU6) and Krupac–RS/FBiH Border (Bogatići) (DMU7), obtained $E = 1.0000$ and were located on the relative efficiency frontier. Trnovo–Dobro Polje (DMU8) was immediately below the frontier, with $E = 0.9760$, while Kula–Krupac (DMU1) ($E = 0.8522$) and Sokolac–Podromanija (DMU9) ($E = 0.7139$) also showed relatively high efficiency scores. By contrast, Podromanija–Rogatica (DMU12) ($E = 0.0895$) and Han Pijesak 2–Sokolac (DMU11) ($E = 0.0897$) recorded the lowest values, followed by Pale 1–Podgrab (DMU5) ($E = 0.1791$) and Podromanija–Sumbulovac (DMU2) ($E = 0.1982$).

The distribution of E values reveals substantial differences in relative safety performance across the analysed road network. These differences are important for interpretation because frontier status is relative rather than absolute. An E value of 1 does not indicate that a road section has inherently better safety performance; it means only that no more favorable weighted relationship between the selected exposure and crash indicators was identified within the analysed set. Similarly, a low E value does not establish the causes of unfavorable safety performance. Instead, it identifies a section whose traffic-exposure and crash-outcome relationship warrants closer examination.

4.2 Results of the Scenario Analysis

A reference scenario with $E^* = 1$ was calculated for the ten sections located below the frontier. Under the adopted radial specification, the adjustment factor corresponds to the initial E value, and the relative change in adverse safety outcomes is expressed as $(1 - E) \times 100\%$. The resulting reference values of the analysed indicators are presented in Table 5. Since Podgrab–RS/FBiH Border (Prača) (DMU6) and Krupac–RS/FBiH Border (Bogatići) (DMU7) were already located on the relative efficiency frontier, no projection was required for these two sections.

Table 5. Reference values of the analysed indicators obtained from the inverse DEA scenario analysis

DMUs	Road Section	E	Required Radial Reduction	FC*	SIC*	LIC*
DMU1	Kula–Krupac	0.8522	14.78%	0.341	3.750	0.852
DMU2	Podromanija–Sumbulovac	0.1982	80.18%	0.040	1.427	0.436
DMU3	Sumbulovac–Ljubogošta	0.3437	65.63%	0.137	1.306	0.619
DMU4	Ljubogošta–Pale 1	0.3751	62.49%	0.075	1.125	0.600
DMU5	Pale 1–Podgrab	0.1791	82.09%	0.143	0.501	0.215
DMU8	Trnovo–Dobro Polje	0.9760	2.40%	0.000	0.781	0.195
DMU9	Sokolac–Podromanija	0.7139	28.61%	0.286	1.428	0.857
DMU10	Vlasenica–Han Pijesak 1	0.3085	69.15%	0.000	0.740	0.247
DMU11	Han Pijesak 2–Sokolac	0.0897	91.03%	0.018	0.466	0.215
DMU12	Podromanija–Rogatica	0.0895	91.05%	0.036	0.931	0.304

Note: DEA = data envelopment analysis; DMUs = decision-making units; E = efficiency score; FC* = reference value of fatal crashes; SIC* = reference value of serious-injury crashes; LIC* = reference value of slight-injury crashes.

The scenario analysis revealed pronounced differences among the sections below the frontier. Trnovo–Dobro Polje (DMU8) required a radial reduction of only 2.40% to reach the selected reference level. The corresponding values for Kula–Krupac (DMU1) and Sokolac–Podromanija (DMU9) were 14.78% and 28.61%, respectively. Considerably larger deviations were obtained for Ljubogošta–Pale 1 (DMU4) (62.49%), Sumbulovac–Ljubogošta (DMU3) (65.63%), and Vlasenica–Han Pijesak 1 (DMU10) (69.15%), followed by Podromanija–Sumbulovac (DMU2) (80.18%) and Pale 1–Podgrab (DMU5) (82.09%). The largest deviations were observed for Han Pijesak 2–Sokolac (DMU11) and Podromanija–Rogatica (DMU12), both exceeding 91%.

These results provide information that cannot be obtained from a simple distinction between sections located on and below the frontier. In particular, sections classified as below the frontier may occupy markedly different relative positions and therefore should not automatically be treated as a homogeneous group in subsequent planning and safety assessment. The radial percentages should not, however, be interpreted as required reductions in actual crash numbers in a future year. They measure the relative distance from the CCR frontier under the specified scenario. Likewise, FC*, SIC*, and LIC* are model-derived reference values rather than forecasts, and there is no assumption that the three crash-severity categories would change proportionally following an actual infrastructure or traffic-management intervention.

4.3 Significance of the Results for Urban and Regional Transport Planning and Road Infrastructure Management

The practical value of the inverse DEA framework lies in moving beyond a general description of road-network conditions toward a more differentiated screening of sections that may require further professional attention. This is

relevant to urban and regional transport planning and road infrastructure management because authorities often need to determine where detailed safety investigations should be undertaken before specific maintenance, reconstruction, or improvement measures are selected. The combination of the E value and the corresponding radial deviation provides an analytical basis for making this initial screening.

Sections with low E values and large deviations from the reference frontier can be identified for earlier and more detailed assessment of factors that are not included directly in the DEA model. These may include road geometry, pavement condition, intersections and access points, speed management, traffic composition, traffic-control devices, and other local characteristics associated with crash occurrence and severity. In this role, the model does not determine which intervention should be implemented. Rather, it helps structure the stage that precedes intervention selection by indicating where more detailed road-safety and infrastructure analyses may be most warranted.

This distinction is particularly relevant when analytical, technical, and financial resources are limited. Treating all sections below the frontier in the same way would overlook the considerable differences observed in their relative deviations. The model can therefore support the sequencing of subsequent investigations and provide additional evidence for maintenance, reconstruction, and infrastructure-improvement planning without replacing the engineering and economic assessments required for individual projects.

Table 6 combines the E values and corresponding radial reductions for all 12 road sections and summarises how the results can be interpreted for transport planning and road infrastructure management.

Table 6. Planning and infrastructure-management interpretation of the relative road-safety results

DMUs	Road Section	E	Required Radial Reduction	Planning and Management
DMU1	Kula–Krupac	0.8522	14.78%	Relatively close to the frontier; monitoring and targeted examination of factors constraining road-safety indicators
DMU2	Podromanija–Sumbulovac	0.1982	80.18%	Large deviation; priority detailed road-safety analysis is justified
DMU3	Sumbulovac–Ljubogošta	0.3437	65.63%	Substantial deviation; pronounced need for detailed road-safety analysis
DMU4	Ljubogošta–Pale 1	0.3751	62.49%	Substantial deviation; pronounced need for detailed road-safety analysis
DMU5	Pale 1–Podgrab	0.1791	82.09%	Large deviation; priority detailed road-safety analysis is justified
DMU6	Podgrab–RS/FBiH Border (Prača)	1.0000	—	Reference section within the analysed set; continuous monitoring of road-safety indicators
DMU7	Krupac–RS/FBiH Border (Bogatići)	1.0000	—	Reference section within the analysed set; continuous monitoring of road-safety indicators
DMU8	Trnovo–Dobro Polje	0.9760	2.40%	Immediately below the frontier; lower relative priority, with continuous monitoring of road-safety indicators
DMU9	Sokolac–Podromanija	0.7139	28.61%	Moderate deviation; targeted monitoring and more detailed analysis subject to available resources
DMU10	Vlasenica–Han Pijesak 1	0.3085	69.15%	Substantial deviation; pronounced need for detailed road-safety analysis
DMU11	Han Pijesak 2–Sokolac	0.0897	91.03%	Largest deviation; strongest need for detailed road-safety and infrastructure assessment
DMU12	Podromanija–Rogatica	0.0895	91.05%	Largest deviation; strongest need for detailed road-safety and infrastructure assessment

Note: DMUs = decision-making units; E = efficiency score; RS = Republic of Srpska; FBiH = Federation of Bosnia and Herzegovina; “—” = not applicable.

Taken together, the results indicate that Han Pijesak 2–Sokolac (DMU11) and Podromanija–Rogatica (DMU12) warrant early attention in a more detailed road-safety assessment because they show the largest relative deviations.

Pale 1–Podgrab (DMU5) and Podromanija–Sumbulovac (DMU2) form the next group, followed by Vlasenica–Han Pijesak 1 (DMU10), Sumbulovac–Ljubogošta (DMU3), and Ljubogošta–Pale 1 (DMU4). This grouping should not be interpreted as a final investment ranking or as a prescribed order of infrastructure interventions. Its purpose is to help direct limited analytical and financial resources toward sections showing the largest relative deviations before specific measures are selected. The model does not identify the causes of crashes and does not prescribe particular engineering measures. Any subsequent intervention decision requires additional evidence on infrastructure condition, traffic characteristics, road-user behavior, spatial context, costs, and expected effects.

4.4 Discussion of the Results

The findings demonstrate the importance of considering traffic exposure together with multiple categories of crash outcomes when comparing road sections. Absolute crash counts alone may disadvantage sections carrying larger traffic volumes, while AADT alone provides no information about the severity or frequency of recorded crashes. DEA places these indicators within a common relative framework and identifies reference units from the observed network. This is consistent with the established role of DEA in road-safety benchmarking, where relative performance measures are used primarily to identify comparatively favorable units and areas in which further examination may be warranted.

The contribution of the present analysis lies in extending this benchmarking logic beyond the classification of sections as being on or below the efficiency frontier. The inverse DEA scenario quantifies the magnitude of the relative deviation associated with each section below the frontier. This distinction has practical relevance because sections sharing the same broad classification may differ substantially in their distance from the reference level.

The contrast between Trnovo–Dobro Polje (DMU8) and the two road sections Han Pijesak 2–Sokolac (DMU11) and Podromanija–Rogatica (DMU12) illustrates this point. Although all three sections were below the frontier, their relative positions were markedly different. Trnovo–Dobro Polje (DMU8) showed a relative deviation of only 2.40% from the selected reference level, whereas Han Pijesak 2–Sokolac (DMU11) and Podromanija–Rogatica (DMU12) showed relative deviations of more than 91%. A binary frontier/non-frontier classification would conceal this difference and provide limited guidance on where further analytical attention may be required. The reference-distance information therefore adds a planning dimension to conventional benchmarking by differentiating the scale of the observed relative shortfall.

At the same time, the two sections with $E = 1$ should not automatically be regarded as examples of generally good road-safety practice. Their position on the frontier is conditional on the analysed sample, the selected indicators, and the adopted model specification. Their geometric, traffic, infrastructure, and spatial characteristics would need to be examined before they could serve as practical benchmarks for other road sections. The same caution applies to sections with low E values: the model identifies an unfavorable relative position but does not establish its underlying cause.

From a planning perspective, this distinction defines the appropriate role of the proposed framework. The results are most useful at an early stage of urban and regional transport planning and infrastructure management, when authorities need to screen a road network and determine where more detailed investigation should be concentrated. Subsequent engineering analysis is required to identify the causes of safety problems and to select appropriate measures. The quantitative framework therefore supports professional decision making by organising comparative evidence, but it does not replace engineering judgement, site-specific assessment, or project-level evaluation.

4.5 Study Limitations and Directions for Future Research

Several limitations should be considered when interpreting the findings. First, the analysis covers only 12 road sections and is based on average annual data for 2019–2023. The resulting frontier is therefore specific to this set of DMUs and should not be interpreted as a universal road-safety benchmark. A larger and more diverse road network could alter both the composition of the frontier and the relative positions of individual sections.

Second, some AADT values for 2022 and 2023 were estimated using linear regression because complete observations were unavailable. Although these estimates were used to complete the traffic-exposure series, uncertainty associated with the estimated values should be considered when interpreting the resulting relative positions. Third, the model includes one exposure indicator and three crash-outcome indicators but does not directly incorporate road geometry, infrastructure condition, weather, spatial characteristics, or road-user behavior. It can therefore identify an unfavorable relative position but cannot determine the factors responsible for that position. Establishing those causes requires detailed crash, infrastructure, and site-specific analysis.

A further limitation arises from the adopted scenario specification. AADT is held constant, while FC, SIC, and LIC are changed proportionally. In actual road systems, different crash-severity categories may respond differently to a given intervention, and infrastructure or traffic-management measures may also affect traffic volumes. The projected values should therefore be interpreted as indicators of relative deviation under the specified model rather than as expected effects of a particular intervention.

Future research could extend the framework to a larger number of road sections and incorporate additional traffic, infrastructure, and spatial indicators where consistent data are available. Testing the stability of the results under alternative model specifications would also provide useful evidence on the robustness of the identified relative positions. In addition, the sections highlighted by the model could be compared with the findings of detailed road-safety inspections and with cost–benefit analyses of planned measures. Such comparisons would provide a stronger test of whether inverse DEA-based screening can reliably support the early stages of maintenance, reconstruction, and investment planning across broader urban and regional road networks.

5 Conclusions

This study examined how an inverse DEA-based framework can support the assessment of relative road-safety performance and the prioritization of road sections within urban and regional transport planning and infrastructure management. By combining the relative position of each section with its distance from a selected reference level, the framework provides more differentiated information than a simple classification of road sections as being on or below the efficiency frontier.

The application to 12 two-lane road sections revealed substantial differences across the analysed network. Two sections formed the relative efficiency frontier, while the remaining sections showed varying degrees of deviation from the reference level. The contrast was particularly pronounced between sections located close to the frontier and those with the least favorable relative positions: one section was only 2.40% below the reference level, whereas the two largest deviations exceeded 91%. These findings demonstrate the value of considering the magnitude of relative deviation when identifying road sections that warrant further investigation.

The main methodological contribution of the study lies in linking DEA-based road-safety benchmarking with a constrained inverse DEA scenario that translates relative performance into explicit reference values. This additional information allows sections below the frontier to be differentiated according to the scale of their relative deviation rather than treated as a single group. In planning terms, the framework provides a transparent screening tool for directing detailed road-safety assessments and supporting the sequencing of subsequent maintenance, reconstruction, and infrastructure-improvement activities.

The practical role of the framework is therefore concentrated at the early stage of decision making. It can help urban and regional transport authorities direct limited analytical, professional, and financial resources toward road sections requiring closer examination before specific interventions are selected. The results do not identify the causes of crashes or determine which engineering measures should be implemented. Such decisions require additional evidence on road geometry, infrastructure condition, traffic characteristics, road-user behavior, spatial context, costs, and expected effects.

The findings should also be interpreted within the boundaries of the adopted model. The efficiency frontier is specific to the analysed road sections and indicators, while the radial inverse DEA scenario provides model-based reference values rather than forecasts of future crash outcomes. The framework should therefore be regarded as a decision-support tool that complements, rather than replaces, detailed engineering assessment and professional judgement. Future research should test the approach on larger and more diverse road networks, incorporate additional infrastructure, traffic, and spatial indicators, and examine whether the sections identified through inverse DEA screening correspond to priorities established through detailed road-safety inspections, cost–benefit analyses, and actual intervention programmes.

Author Contributions

Conceptualization, M.Đ. and M.S.; methodology, M.Đ.; validation, M.S. and R.U.-V.; formal analysis, M.Đ.; data curation, M.Đ.; writing—original draft preparation, M.Đ. and M.S.; writing—review and editing, R.U.-V. and B.G.; supervision, B.G. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] R. D. Banker, A. Charnes, and W. W. Cooper, “Some models for estimating technical and scale inefficiencies in data envelopment analysis,” *Manage. Sci.*, vol. 30, no. 9, pp. 1078–1092, 1984. <https://doi.org/10.1287/mnsc.30.9.1078>
- [2] Q. Wei, J. Zhang, and X. Zhang, “An inverse DEA model for inputs/outputs estimate,” *Eur. J. Oper. Res.*, vol. 121, no. 1, pp. 151–163, 2000. [https://doi.org/10.1016/S0377-2217\(99\)00007-7](https://doi.org/10.1016/S0377-2217(99)00007-7)

- [3] A. Charnes, W. W. Cooper, and E. Rhodes, "Measuring the efficiency of decision-making units," *Eur. J. Oper. Res.*, vol. 2, no. 6, pp. 429–444, 1978. [https://doi.org/10.1016/0377-2217\(78\)90138-8](https://doi.org/10.1016/0377-2217(78)90138-8)
- [4] J. Odeck, "Assessing the relative efficiency and productivity growth of vehicle inspection services: An application of DEA and Malmquist indices," *Eur. J. Oper. Res.*, vol. 126, no. 3, pp. 501–514, 2000. [https://doi.org/10.1016/S0377-2217\(99\)00305-7](https://doi.org/10.1016/S0377-2217(99)00305-7)
- [5] J. Odeck, "Identifying traffic safety best practice: An application of DEA and Malmquist indices," *Omega*, vol. 34, no. 1, pp. 28–40, 2006. <https://doi.org/10.1016/j.omega.2004.07.017>
- [6] M. E. Ozbek, J. M. de la Garza, and K. Triantis, "Data and modeling issues faced during the efficiency measurement of road maintenance using data envelopment analysis," *J. Infrastruct. Syst.*, vol. 16, no. 1, pp. 21–30, 2010. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2010\)16:1\(21\)](https://doi.org/10.1061/(ASCE)1076-0342(2010)16:1(21))
- [7] G. Fancello, B. Uccheddu, and P. Fadda, "Data envelopment analysis (D.E.A.) for urban road system performance assessment," *Procedia Soc. Behav. Sci.*, vol. 111, pp. 780–789, 2014. <https://doi.org/10.1016/j.sbspro.2014.01.112>
- [8] E. Hermans, T. Brijs, G. Wets, and K. Vanhoof, "Benchmarking road safety: Lessons to learn from a data envelopment analysis," *Accid. Anal. Prev.*, vol. 41, no. 1, pp. 174–182, 2009. <https://doi.org/10.1016/j.aap.2008.10.010>
- [9] Y. Shen, E. Hermans, T. Brijs, G. Wets, and K. Vanhoof, "Road safety risk evaluation and target setting using data envelopment analysis and its extensions," *Accid. Anal. Prev.*, vol. 48, pp. 430–441, 2012. <https://doi.org/10.1016/j.aap.2012.02.020>
- [10] A. Van Espen, "Evaluating traffic safety performance of countries using data envelopment analysis and accident prediction models," Ph.D. dissertation, University of British Columbia, Vancouver, Canada, 2013.
- [11] P. Nikolaou and L. Dimitriou, "Evaluation of road safety policies performance across europe: Results from benchmark analysis for a decade," *Transp. Res. Part A Policy Pract.*, vol. 116, pp. 232–246, 2018. <https://doi.org/10.1016/j.tra.2018.06.026>
- [12] Y. Shen, E. Hermans, Q. Bao, T. Brijs, and G. Wets, "Towards better road safety management: Lessons learned from international benchmarking," *Accid. Anal. Prev.*, vol. 138, p. 105484, 2020. <https://doi.org/10.1016/j.aap.2020.105484>
- [13] D. Alper, Z. Sinuany-Stern, and D. Shinar, "Evaluating the efficiency of local municipalities in providing traffic safety using the data envelopment analysis," *Accid. Anal. Prev.*, vol. 78, pp. 39–50, 2015. <https://doi.org/10.1016/j.aap.2015.02.014>
- [14] B. Antić, M. Grdinić, D. Pešić, and V. Pajković, "Benchmarking of the road safety performance among the regions by using DEA," *Transp. Res. Procedia*, vol. 45, pp. 78–86, 2020. <https://doi.org/10.1016/j.trpro.2020.02.065>
- [15] G. Egilmez and D. McAvoy, "Benchmarking road safety of U.S. states: A DEA-based Malmquist productivity index approach," *Accid. Anal. Prev.*, vol. 53, pp. 55–64, 2013. <https://doi.org/10.1016/j.aap.2012.12.038>
- [16] J. T. Bastos, Y. Shen, E. Hermans, T. Brijs, G. Wets, and A. C. P. Ferraz, "Traffic fatality indicators in Brazil: State diagnosis based on data envelopment analysis research," *Accid. Anal. Prev.*, vol. 81, pp. 61–73, 2015. <https://doi.org/10.1016/j.aap.2015.01.024>
- [17] C. Xu, C. Feng, and D. Hu, "Evaluation of the performance of provincial road traffic safety in China based on two-stage DEA model," in *2019 International Conference on Industrial Engineering and Systems Management (IESM)*, Shanghai, China, 2019, pp. 1–6. <https://doi.org/10.1109/IESM45758.2019.8948168>
- [18] J. Mitrović Simić, Ž. Stević, E. K. Zavadskas, V. Bogdanović, M. Subotić, and A. Mardani, "A novel CRITIC-fuzzy FUCOM-DEA-fuzzy MARCOS model for safety evaluation of road sections based on geometric parameters of road," *Symmetry*, vol. 12, no. 12, p. 2006, 2020. <https://doi.org/10.3390/sym12122006>
- [19] A. Sadeghi and A. Mohammadzadeh Moghaddam, "Uncertainty-based prioritization of road safety projects: An application of data envelopment analysis," *Transp. Policy*, vol. 52, pp. 28–36, 2016. <https://doi.org/10.1016/j.tranpol.2016.07.003>
- [20] S. Cafiso, A. Di Graziano, G. Di Silvestro, G. La Cava, and B. Persaud, "Development of comprehensive accident models for two-lane rural highways using exposure, geometry, consistency and context variables," *Accid. Anal. Prev.*, vol. 42, no. 4, pp. 1072–1079, 2010. <https://doi.org/10.1016/j.aap.2009.12.015>
- [21] A. Montella, "A comparative analysis of hotspot identification methods," *Accid. Anal. Prev.*, vol. 42, no. 2, pp. 571–581, 2010. <https://doi.org/10.1016/j.aap.2009.09.025>
- [22] M. Damjanović, Ž. Stević, D. Stanimirović, I. Tanackov, and D. Marinković, "Impact of the number of vehicles on traffic safety: Multiphase modeling," *Facta Univ. Ser. Mech. Eng.*, vol. 20, no. 1, pp. 177–197, 2022. <https://doi.org/10.22190/FUME220215012D>
- [23] Ž. Stević, D. K. Das, and M. Kopic, "A novel multiphase model for traffic safety evaluation: A case study of South Africa," *Math. Probl. Eng.*, vol. 2021, p. 5584599, 2021. <https://doi.org/10.1155/2021/5584599>

- [24] S. Gattoufi, G. R. Amin, and A. Emrouznejad, "A new Inverse DEA method for merging banks," *IMA J. Manag. Math.*, vol. 25, no. 1, pp. 73–87, 2014. <https://doi.org/10.1093/imaman/dps027>
- [25] A. Emrouznejad, G. R. Amin, M. Ghiyasi, and M. Michali, "A review of inverse data envelopment analysis: Origins, development and future directions," *IMA J. Manag. Math.*, vol. 34, no. 3, pp. 421–440, 2023. <https://doi.org/10.1093/imaman/dpad006>